

ECE 635 Homework #3 Solutions

Fall 2025

1. (a) $f = 3.18 \text{ GHz}$

(b) 5 V/m

(c) $-z$ direction

(d) $1.73 \times 10^8 \text{ m/s}$

(e) 5.4 cm

(f) 217.66 S

(g) 115.47 rad/m

2. 3.67

3. Use the method in the notes for the electric field Helmholtz equation.

4. For each of the following problems a complete set of steps leading to the polarization equation and the polarization angle is expected.

(a) RHCP ; $\psi = \omega t$

(b) RHCP ; $\psi = \omega t$

(c) RHEP ; $\psi =$ a ratio of E_x over E_z

(d) Straight line ; $\psi = 63.4^\circ$